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Objectives of the Project

The main objectives of this project are:

- To analyze and document the requirements of the SpiceSync food ordering system.
- To identify the functional and non-functional requirements of the system.
- To design the overall structure and workflow of the system using UML diagrams.
- To create a UML Class Diagram representing system classes, attributes, methods, and relationships.
- To create an Activity/Swimlane Diagram representing the complete workflow of the system.
- To understand and represent the interaction between different system actors such as customers, restaurants, delivery riders, and administrators.
- To study and apply the MVC (Model-View-Controller) architectural pattern for system organization.
- To improve understanding of software engineering and system design concepts.
- To develop documentation and modeling skills for real-world software systems.
- To present a clear and organized software design for the proposed SpiceSync system.

Introduction

Food delivery applications have become an important part of modern life. Customers expect fast ordering, personalized meal options, secure payments, and real-time tracking services. Traditional food delivery systems often lack regional customization, multilingual support, and advanced engagement features.

SpiceSync is developed to solve these problems by providing a smart and culturally customized food ordering platform. The system focuses on Pakistani users by supporting local spice preferences, halal transparency, local payment methods, and Urdu language support.

The project also includes advanced features such as AI-based recommendations, loyalty rewards, group ordering, subscription plans, and voice-assisted ordering to improve user experience and customer satisfaction.

Requirements

Functional Requirements:

FR-1 User Registration and Login

The system shall allow users to:

- Register using email and password

- Login using email/password
- Login using Google or Facebook accounts
- Reset forgotten passwords
- Verify email during account creation

FR-2 User Profile Management

The system shall allow users to:

- Edit personal information
- Save delivery addresses
- Save preferred language
- Save favorite meals and spice preferences

FR-3 Meal Browsing and Search

The system shall allow users to:

- Browse food categories
- Search meals by name
- View meal details including price and ingredient

FR-4 AI-Based Meal Recommendations

The system shall recommend meals based on:

- User preferences
- Previous orders
- Spice level choices
- Regional food interests

FR-5 Regional Spice Customization

The system shall allow users to customize meals according to regional spice levels such as:

- Punjabi spicy
- Karachi spicy
- Peshawari style
- Islamabad mil

FR-6 Cart Management

The system shall allow users to:

- Add meals to cart

- Remove meals from cart
- Update meal quantity
- View total price before checkout

FR-7 Order Placement

The system shall allow users to:

- Place delivery orders
- Place pickup orders
- Confirm order details before checkout

FR-8 Multiple Payment Options

The system shall support:

- JazzCash
- Easypaisa
- Cash on Delivery
- Debit/Credit cards

FR-9 Real-Time Order Tracking

The system shall:

- Display order status updates
- Show estimated delivery time
- Provide live tracking information

FR-10 Multilingual Support

The system shall support:

- English language
- Urdu language

The user shall be able to switch languages easily.

FR-11 Halal Certification and Nutritional Information

The system shall:

- Display halal certification information
- Show ingredients and calorie details for meals

FR-12 Loyalty Reward System

The system shall:

- Award Spice Points on purchases
- Allow users to redeem reward points for discounts

FR-13 Group Ordering

The system shall allow users to:

- Create group orders
- Add multiple meals in one shared order
- Receive bulk discounts for large orders

FR-14 Subscription Plans

The system shall allow users to:

- Subscribe to weekly or monthly meal plans
- Manage office lunch subscriptions

FR-15 Recipe Contest Feature

The system shall allow users to:

- Submit custom recipes
- Participate in recipe contests
- Vote on recipes

FR-16 Feedback and Rating System

The system shall allow users to:

- Rate meals and restaurants
- Submit feedback and reviews after delivery

FR-17 Voice-Assisted Ordering (SpiceBot)

The system shall support:

- Urdu voice commands
- English voice commands

The system shall allow users to place orders using voice assistance.

FR-18 Restaurant Management

The system shall allow restaurant partners to:

- Manage menus
- Update meal prices
- View customer orders
- Update order preparation status

FR-19 Delivery Management

The system shall:

- Assign delivery riders
- Manage delivery status
- Optimize delivery operations

FR-20 Admin Dashboard and System Management

The system shall allow administrators to:

- Manage users and restaurants
- Monitor orders and payments
- Manage promotions and feedback
- Generate reports and analytics

Non-Functional Requirements:

NFR-1 Performance

The system shall:

- Load pages within a few seconds under normal internet conditions.
- Handle multiple users and orders simultaneously.
- Provide smooth performance during peak hours.

NFR-2 Reliability

The system shall:

- Operate continuously with minimal downtime.
- Ensure accurate order processing and tracking.
- Recover safely from system failures or crashes.

NFR-3 Availability

The system shall be available 24/7 for users, restaurants, delivery riders, and administrators.

NFR-4 Security

The system shall:

- Protect user personal and payment data.
- Encrypt passwords and sensitive information.
- Prevent unauthorized system access.
- Provide secure payment transactions.

NFR-5 Usability

The system shall:

- Provide a simple and user-friendly interface.
- Support easy navigation for all user types.
- Allow users to complete ordering tasks easily.

NFR-6 Accessibility

The system shall:

- Support Urdu and English languages.
- Support voice-assisted ordering.
- Maintain readable text and clear interface elements.

NFR-7 Scalability

The system shall:

- Support increasing numbers of users, restaurants, and orders.
- Allow future addition of new cities and restaurants.

NFR-8 Compatibility

The system shall:

- Work on Android and iOS devices.
- Support major web browsers such as Chrome, Firefox, and Safari.

NFR-9 Maintainability

The system shall:

- Allow easy system updates and feature modifications.
- Maintain organized and manageable code structure.

NFR-10 Database Management

The system shall:

- Store user and order data securely.
- Maintain backup and recovery mechanisms.
- Ensure data consistency and integrity.

NFR-11 Response Time

The system shall:

- Respond quickly to user actions.
- Provide real-time updates for order tracking and notifications.

NFR-12 Network Support

The system shall:

- Function properly under low-bandwidth internet conditions.
- Handle temporary internet interruptions gracefully.

NFR-13 Localization

The system shall:

- Support Pakistani regional preferences and cultural requirements.
- Support local payment systems and halal transparency.

NFR-14 Portability

The system shall:

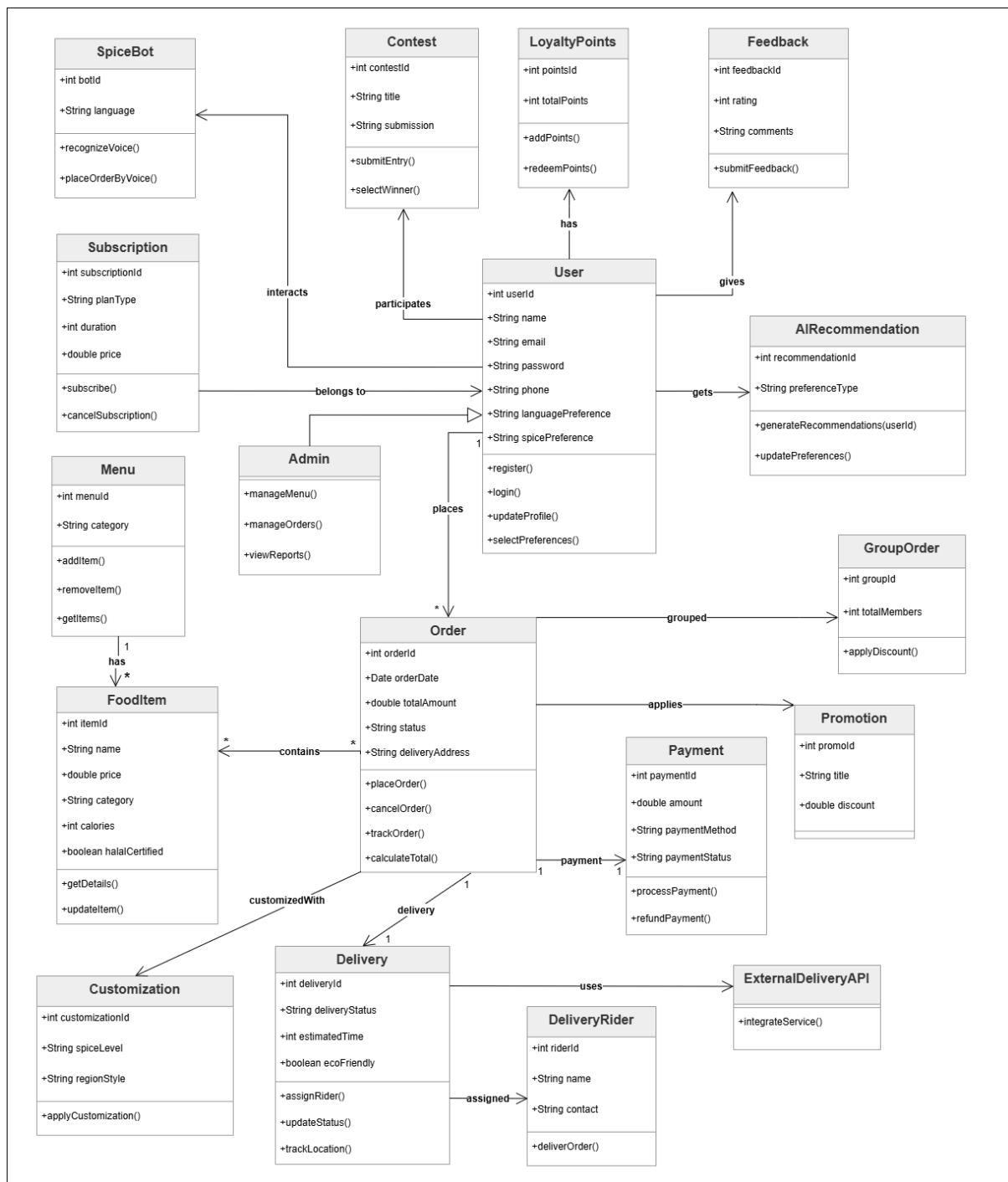
- Be deployable on different devices and operating systems without major changes.

NFR-15 Data Integrity

The system shall:

- Ensure accurate storage and retrieval of orders, payments, and customer information.
- Prevent duplication or corruption of records.

UML Digram



This is a UML Class Diagram for the SpiceSync system.

The diagram represents all the major classes used in the application along with their attributes, methods, and relationships. It explains how different parts of the system interact with each other to manage food ordering, customization, payments, delivery, rewards, and administration.

Classes in the Diagram

Class Name	Description
User	Stores customer information such as name, email, password, preferred language, and spice preference.
Menu	Contains food item details including item name, price, calories, and halal status.
Order	Handles customer orders including placing, canceling, tracking, and calculating total price.
Payment	Manages payment processing and refund functionality.
Delivery	Handles delivery operations such as assigning riders, updating delivery status, and tracking location.
Customization	Manages spice levels and regional flavor selections such as Punjabi or Peshawari flavors.
Subscription	Handles office lunch and monthly tiffin subscription plans.
LoyaltyPoints	Manages Spice Points including adding and redeeming reward points.
Feedback	Stores customer ratings and reviews for food and services.
Contest	Handles recipe contests including entry submission and winner selection.
GroupOrder	Supports group ordering and bulk discount management.
Promotions	Manages seasonal and festive offers such as Eid, Ramadan, and Independence Day deals.
SpiceBot	Voice assistant used for voice ordering and speech recognition.
Admin	Allows administrators to manage menu items by adding, removing, or updating them.
ExternalDeliveryAPI	Integrates external delivery services such as Foodpanda and Cheetay.

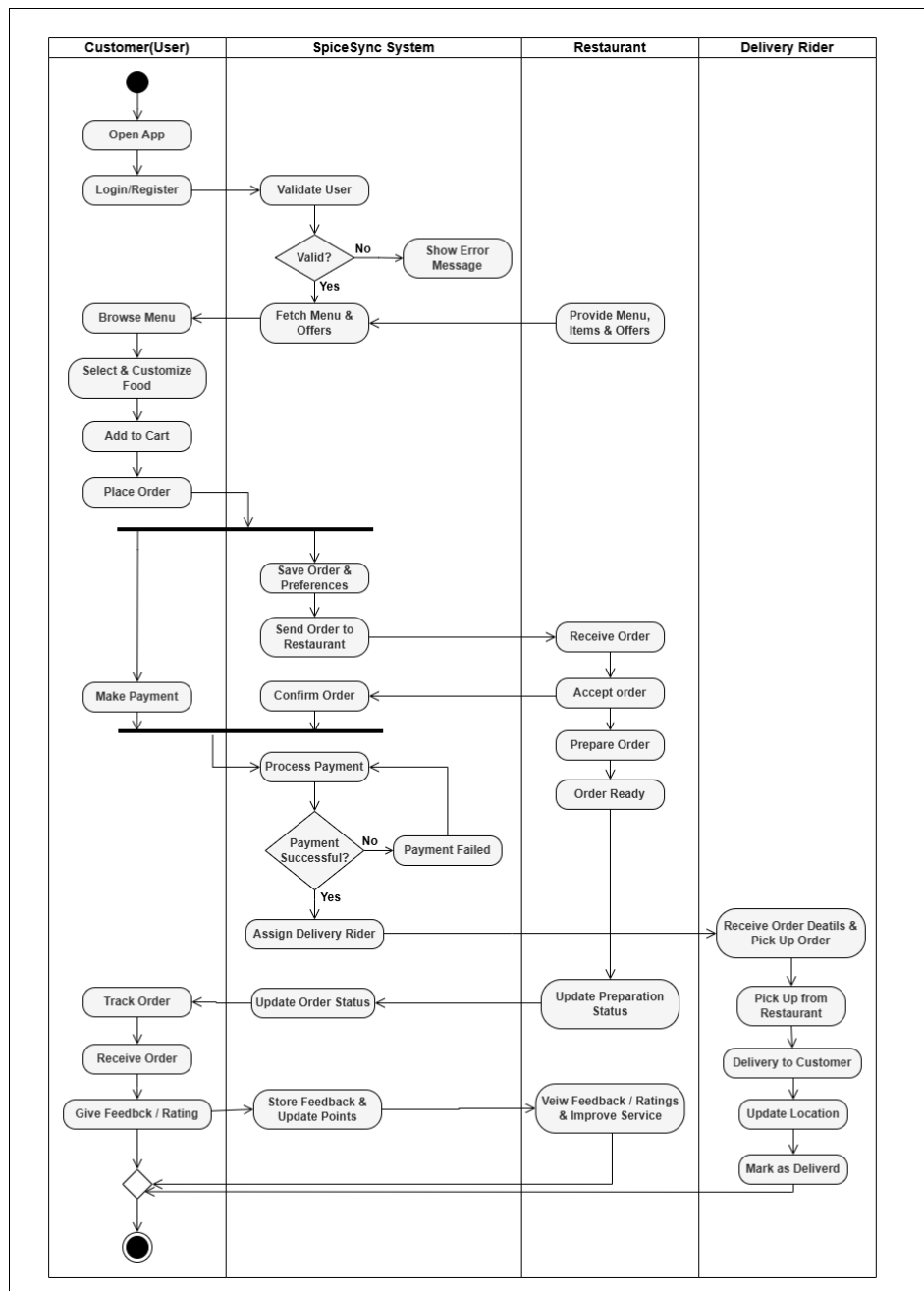
Relationships Between Classes

Relationship	Description
User → Order	One user can place multiple orders.
Order → Payment	Each order has one payment associated with it.
Order → Delivery	Each order is connected to one delivery process.
Order → Menu	One order can contain multiple menu items.
User → Subscription	A user can subscribe to a lunch or tiffin plan.
User → LoyaltyPoints	Each user has one loyalty points account.
User → Feedback	One user can submit multiple feedbacks and reviews.
Admin → Menu	The admin manages all menu items in the system.

Category-Wise Classification of Classes

Category	Classes Included
User Management	User, Order, Payment, Delivery
Food Management	Menu, Customization
Rewards and Engagement	LoyaltyPoints, Feedback, Contest, Promotions
Special Features	Subscription, GroupOrder, SpiceBot
Administration and Integration	Admin, ExternalDeliveryAPI

SwinLane



This is a UML Activity Diagram, also known as a Flow Diagram or Swimlane Diagram.

It is used to represent:

- The sequence of activities in the system
- The interaction between different actors
- Decision-making processes
- Workflow and process execution
- Parallel activities performed by multiple entities

The SpiceSync Activity Diagram provides a complete representation of the food ordering workflow within the system. It demonstrates how customers interact with the application, how payments and deliveries are managed, and how feedback and loyalty systems are integrated. The diagram clearly illustrates the responsibilities of each actor and ensures smooth coordination between the customer, system, restaurant, and delivery rider.

Main Actors in the Diagram

Actor	Description
Customer (User)	The person who places food orders through the SpiceSync application.
SpiceSync System	The application that manages ordering, payment, tracking, and processing.
Restaurant	The restaurant that prepares the customer’s food order.
Delivery Rider	The rider responsible for delivering food to the customer.

Complete Workflow of the System

Phase 1: User Login and Authentication

Step	Activity	Decision
1	Customer opens the SpiceSync application.	-
2	Customer logs in or creates a new account.	-
3	System validates user credentials.	Valid or Invalid
4	If credentials are invalid, the system displays an error message and asks the user to try again.	-
5	If credentials are valid, the system fetches menu items and promotional offers.	-

Phase 2: Food Ordering Process

Step	Activity
6	The system displays available menu items and offers.
7	The customer selects food items and customizes spice levels or flavors.
8	The selected items are added to the cart.
9	The customer places the order.
10	The system stores order details and customer preferences.
11	The order request is sent to the restaurant.

Phase 3: Payment Processing

Step	Activity	Decision
12	Customer proceeds with payment.	-
13	The system processes and verifies the payment.	Payment Success or Failure
14	If payment fails, the system shows an error message and asks the customer to retry payment.	-
15	If payment is successful, the system proceeds to delivery management.	-

Phase 4: Delivery Process

Step	Activity
16	The system assigns a delivery rider.
17	The customer tracks the order in real time.
18	The restaurant prepares the food and hands it to the rider.
19	The rider picks up the order from the restaurant.
20	The rider delivers the order to the customer.
21	The rider continuously updates the delivery location.
22	The rider marks the order as delivered.
23	The customer receives the food order.

Phase 5: Feedback and Loyalty Points

Step	Activity
24	Customer submits feedback and ratings for the order.
25	The system stores feedback and updates customer loyalty points.

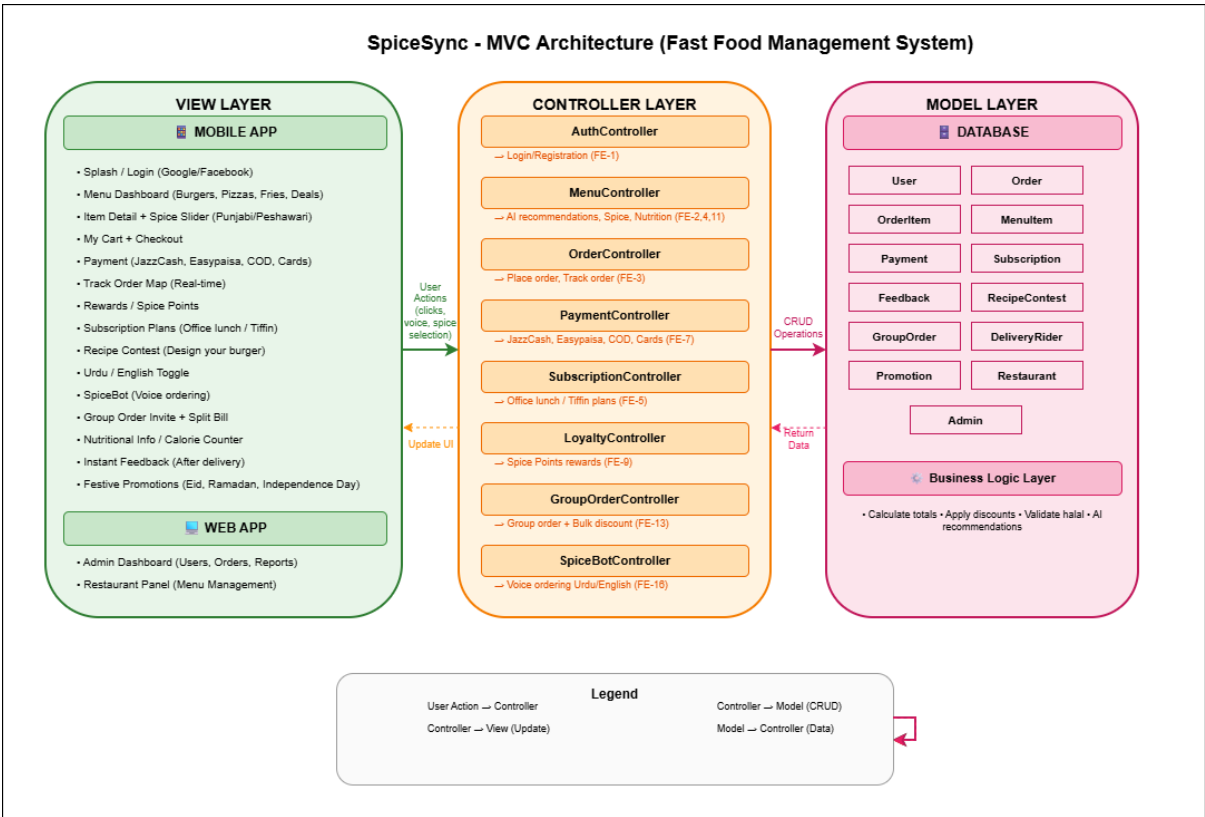
Delivery Options in the Diagram

Delivery Option	Description
Home Delivery	Food is delivered from the restaurant to the customer through a delivery rider.
Restaurant Pickup	The customer directly collects the order from the restaurant.

Summary Flow of the System

Open App → Login/Register → Browse Menu → Customize Food → Add to Cart → Place Order → Make Payment → Payment Verification → Assign Rider → Track Order → Receive Food → Submit Feedback → Earn Loyalty Points

MVC Architecture



MVC (Model-View-Controller)

MVC (Model-View-Controller) is a design pattern that splits an application into three connected parts. This makes the code organized, easy to maintain, and simple to update.

Component Role

View: User interface (what user sees)

Controller: Business logic (handles requests)

Model: Data and database (stores information)

1. View Layer (User Interface)

The View displays data to users and captures their input.

Mobile App Screens:

- Login/Signup (Google/Facebook)
- Menu Dashboard
- Item Detail + Spice Slider (Punjabi/Peshawari)
- Cart + Checkout
- Payment (JazzCash, Easypaisa, COD)
- Track Order Map
- Rewards / Spice Points
- Subscription Plans
- Recipe Contest
- Urdu/English Toggle
- SpiceBot Voice
- Group Order + Split Bill
- Nutritional Info
- Instant Feedback
- Festive Promotions

Web App Screens:

- Admin Dashboard
- Restaurant Panel
- Delivery Panel

2. Controller Layer (Business Logic)

The Controller receives user requests, calls the Model, and updates the View.

Controller Features

- AuthController Login/Registration (FE-1)
- MenuController AI recommendations, Spice, Nutrition (FE-2,4,11)
- OrderController Place order, Track order (FE-3)
- PaymentController JazzCash, Easypaisa, COD (FE-7)
- LoyaltyController Spice Points (FE-9)
- GroupOrderController Group order + Bulk discount (FE-13)
- SpiceBotController Voice ordering (FE-16)

3. Model Layer (Data & Database)

The Model manages data storage and business logic.

Database Tables:

· User, Order, OrderItem, MenuItem, Payment, Subscription, Feedback, RecipeContest, GroupOrder, DeliveryRider, Promotion, Restaurant, Admin

Business Logic:

- calculateTotal() → Order amount
- applyBulkDiscount() → 10% off on 5+ meals
- validateHalal() → Halal check
- getRecommendations() → AI suggestions
- calculatePoints() → Spice Points
- getNutrition() → Calorie counter

4. How They Interact (The Flow)

User adds item to cart:

Step Action

1. User clicks "Add to Cart" → View captures request
2. View → Controller sends item details
3. Controller → Model saves to database
4. Model → Controller returns success + new total
5. Controller → View updates cart badge
6. View shows updated cart to user

5. Why Use MVC?

Benefit Explanation

- Organized : Each layer has its own folder and responsibility
- Maintainable : Change one layer without breaking others
- Reusable : Same Model can work with different Views
- Testable : Each layer can be tested separately
- Secure : Sensitive data stays in Model, away from View

Conclusion

SpiceSync is a modern food ordering and delivery platform designed to improve the overall customer experience through personalization, convenience, and smart technology. The system combines essential food delivery features with advanced functionalities such as AI-

based recommendations, regional spice customization, multilingual support, loyalty rewards, and voice-assisted ordering.

The project successfully demonstrates the use of software engineering concepts including functional and non-functional requirements, UML class diagrams, activity diagrams, swimlane workflows, and MVC architecture. These components help ensure that the system is organized, scalable, secure, and easy to maintain.

Overall, SpiceSync provides an efficient solution for customers, restaurants, delivery riders, and administrators while supporting future enhancements and scalability.